

CHAPTER 8

TENDON-DRIVEN MANIPULATORS

8.1 INTRODUCTION

The kinematic structure of a robot manipulator often takes the form of an open-loop chain. An open-loop manipulator is mechanically simple and easy to construct. Theoretically, one actuator can be mounted on each link to drive the next link via a speed reduction unit. Using this approach, however, actuators and speed reducers installed on the distal end become the load for actuators installed on the proximal end of a manipulator, resulting in a bulky and heavy system. To reduce the size and inertia of a robot manipulator, it is often necessary to introduce some mechanical transmission system that permits actuators to be located remotely away from the joints.

Various mechanical transmission systems, including gear trains, tendon-and-pulley trains, tie-rod linkages, and so on, may be incorporated. The type of transmission system selected depends on the application and other design considerations. Generally, the power-to-weight ratio must be optimized, backlash and vibration minimized, and friction reduced. The application of gear trains as mechanical transmission mechanisms for robot wrists was discussed in Chapter 7. In this chapter we concentrate on the kinematics of tendon drives. The term *tendon* is widely used to imply belts, cables, or similar constructions. The main advantages of using tendon drives are: (1) All actuators can be installed on the fixed base, resulting in a compact and lightweight design; and (2) a well-designed tendon transmission system has little backlash. These merits have made tendons better suited than other mechanical transmission systems in applications such as dextrous hands, where the re-

quirements of small volume, light weight, and high speed are the most important factors. The main disadvantages of introducing a tendon transmission system are the extra cost of additional mechanical components and the possibility of creating some drawbacks, such as friction and vibration in the system.

In this chapter we address the fundamental mechanics of tendon-driven manipulators. Basic issues, including the structural characteristics, kinematics, and statics of tendon-driven manipulators, are studied.

8.2 CLASSIFICATION OF TENDON-DRIVEN MANIPULATORS

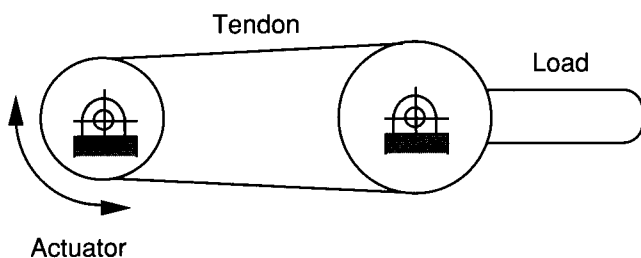
Tendon drives can generally be classified into two categories: (1) *closed-loop tendon drives* and (2) *open-ended tendon drives*. In what follows, we describe each of them in turn.

8.2.1 Closed-Loop Tendon Drives

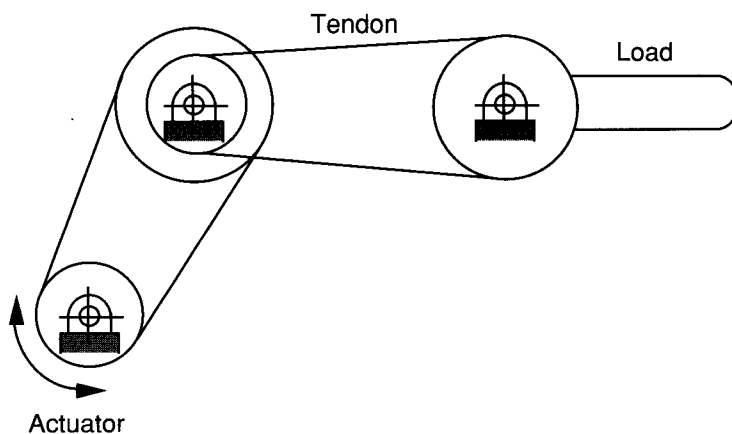
Perhaps the most commonly known tendon-driven manipulating system is the *closed-loop tendon drive*, used in a variety of industrial machinery. Recently, these drives have also been incorporated in robot manipulators (Ali et al., 1993; Cannon and Schmitz, 1984; Hollars and Cannon, 1985; Melchiorri and Vassura, 1992; Okada, 1977; Sugano and Kato, 1987). This type of transmission system uses one motor to drive a closed-loop belt in both directions. Figure 8.1a shows a 1-dof mechanism driven by a rotary actuator through a closed-loop tendon. Power transmission in a closed-loop tendon transmission system usually relies on friction generated between pulleys and belts. To increase the efficiency, toothed belts known as the *timing belt* or *chain-and-sprocket devices* can be used. Closed-loop tendons can also be connected in series to form a multistage transmission system, as shown in Fig. 8.1b. In a closed-loop drive, a portion of each tendon will be under high tension while the other portion is under little tension. Although torque can be transmitted in both directions, pretension is often necessary to prevent tendons from slackening. Pretension, however, can introduce significant amount of friction and apparent backlash due to the elastic effect of tendons. Because a closed-loop belt can be driven in both directions, the number of actuators needed is equal to the number of degrees of freedom of a manipulator.

8.2.2 Open-Ended Tendon Drives

Another form of tendon transmission is called an *open-ended tendon drive*. Let n be the degrees of freedom and m the number of control tendons in an open-ended tendon drive. We call such a manipulator an $n \times m$ manipulator.



(a) Single-stage closed-loop tendon drive.



(b) Double-stage closed-loop tendon drive.

FIGURE 8.1. Closed-loop tendon drives.

For example, Fig. 8.2 shows a 2×3 tendon-driven manipulator. In an open-ended tendon transmission system, one end of each tendon is attached to a moving link while the other end is pulled by an actuator. Force is transmitted by pulling of the tendons. A unique feature associated with open-ended tendon drives is that tendons can exert tension but not compression. That is, actuator forces must be applied in a unidirectional sense. Morecki et al. (1980) showed that m should be greater than n to achieve independent control of all the degrees of freedom of a manipulator. Using this criterion, we can classify open-ended tendon-driven manipulators as follows.

(a) $m < n$. If the number of tendons is less than the number of degrees of freedom, the motion of a manipulator cannot be controlled at will. We call this type of manipulator an *insufficiently actuated manipulator*. An insufficiently

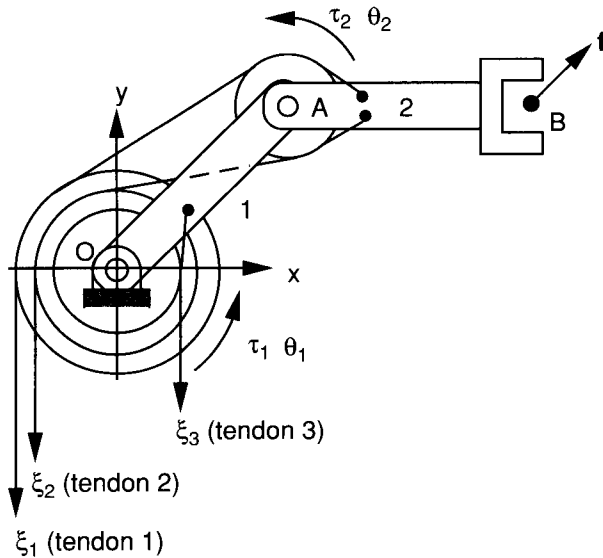


FIGURE 8.2. A 2-dof manipulator driven by three tendons.

actuated manipulator usually relies on mechanical constraints or structural characteristics to control its posture. Hirose and Umetani (1978, 1979) developed a soft gripper. The gripper consists of three multi-dof fingers. Each is controlled by only one grip tendon and one release tendon. The diameter of the grip pulley becomes smaller from its base joint toward the tip joint, while the diameter of the release pulley is uniform along its entire length. Although the joints cannot be controlled independently, the fingers can conform to an object of random shape with a uniform grasping force. Rovetta (1981) constructed a mechanical hand with two fingers and a palm. Each finger has four joints and is pulled by a single tendon against a spring-loaded palm. Insufficiently actuated manipulators have also been configured into *differential mechanisms* (Hirose, 1986).

(b) $m = n$. When the number of tendons is equal to the number of degrees of freedom, the manipulator is still an insufficiently actuated mechanism. This type of manipulator can often be found in hoisting cranes and elevators for raising, shifting, and lowering heavy objects. For such applications, tendons are usually designed to pull against the gravitational force. A robotic crane system utilizing the Stewart platform architecture was recently developed by Albus et al. (1992). In their design, six cables are used as parallel links to manipulate the location of a suspended moving platform, as shown in Fig. 8.3. Although the number of actuators is less than $n + 1$, complete control of

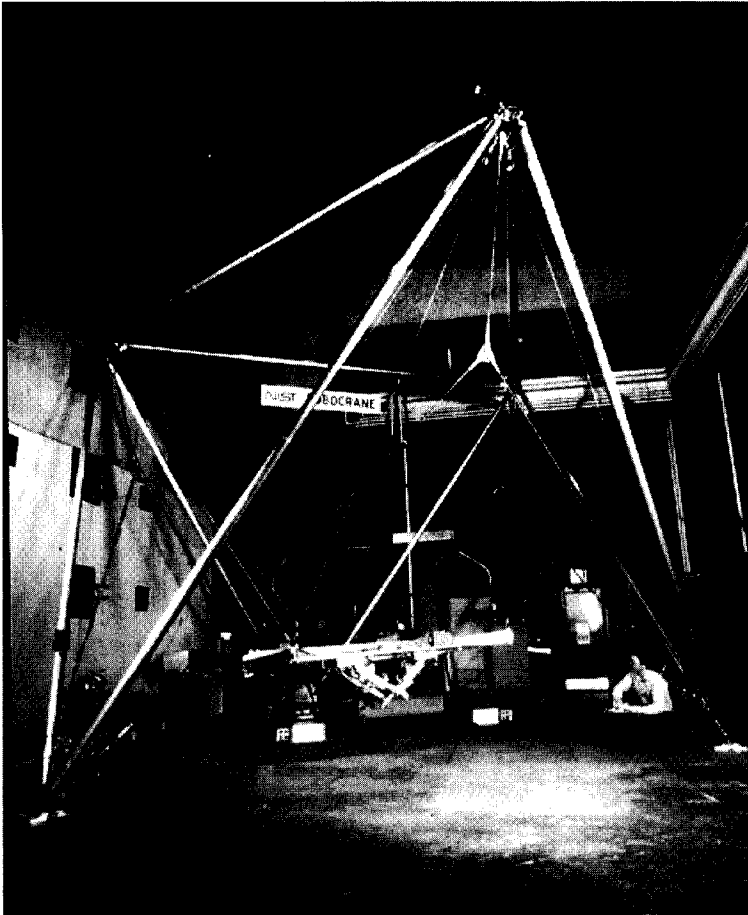


FIGURE 8.3. NIST RoboCrane. (Courtesy of NIST, Gaithersburg, Maryland.)

the end-effector location is possible. In fact, the gravitational force has been employed as the $(n + 1)$ th control force.

Another way of controlling an n -dof manipulator with n tendons is to let the actuators work against spring-loaded tendons or joints. For example, Rovetta (1977) built a mechanical gripper in which return springs were installed in the joints to serve as a bias torque. This approach, however, prohibits the system from fine manipulation and force control since the spring may exhibit some nonlinearity and cause asymmetric responses.

(c) $m > n$. If the number of tendons exceeds the number of degrees of freedom, the manipulator can be controlled at will. This is by far the most popular arrangement because it allows users complete control of the degrees

of freedom. We call this type of manipulator a *sufficiently actuated manipulator*. For example, Morecki et al. (1980) discussed some of the problems encountered in the design of their anthropomorphic two-handed manipulator and Salisbury designed a three-fingered hand known as the *Stanford/JPL hand* (Salisbury, 1982; Rouff and Salisbury, 1990). In the Stanford/JPL hand, each finger has three articulation points and is controlled by four tendons. The kinematics and statics of the transmission system were studied in detail by Salisbury and Craig (1982) and Salisbury and Roth (1983). As shown in Fig. 8.4, Jacobsen et al. (1984, 1986, 1989) developed a four-fingered hand



FIGURE 8.4. Utah/MIT dextrous hand. (Courtesy of SARCOS, Salt Lake City, Utah.)

called the *Utah/MIT hand*. Each finger of the Utah/MIT hand has four articulation points and is controlled by eight tendons. Many other tendon-driven manipulators are discussed in Pham and Heginbotham (1986).

8.3 PLANAR SCHEMATIC REPRESENTATION

In this section we present a planar schematic representation of the kinematic structure of tendon-driven manipulators. To reduce the scope of study, we consider only those mechanisms that obey the following assumptions:

1. The manipulator is an articulated type (i.e., after removal of tendons and pulleys, the mechanism becomes an open-loop chain).
2. Each tendon is routed from the base link to one of the moving links over pulleys mounted on the joint axes of an open-loop chain in a consecutive manner.
3. All tendons are under tension, and the amount of stretch in each tendon is negligible.
4. No slippage occurs between pulleys and tendons.

Figure 8.5 shows a functional schematic diagram of the Stanford/JPL finger (Mason and Salisbury, 1985). For the purpose of illustration, the fixed base is denoted by link 0, the moving links and joints are numbered sequentially from 1 to 3, and a positive direction of rotation is assigned to each joint axis, as shown in the figure. In the Stanford/JPL finger, the first joint axis, z_1 , is fixed to the base link; the second joint axis, z_2 , is perpendicular to the first; and the third joint axis, z_3 , is parallel to the second. Pulley 2 is attached to link 2, and pulley 3 is attached to link 3. Pulleys 4, 5, 6, and 7 are free to rotate about the first joint axis, pulleys 2 and 8 are free to rotate about the second joint axis, and pulley 3 is free to rotate about the third joint axis. The first tendon, s_1 , connects pulley 4 to 8 and then 3; the second tendon, s_2 , connects pulley 5 to 8 and then 3; the third tendon, s_3 , connects pulley 6 to 2; and the fourth tendon, s_4 , connects pulley 7 to 2. Overall, the mechanism consists of nine links and four control tendons.

The tendon routings in such a functional schematic diagram can sometimes be confusing due to the nature of spatial construction. For convenience, a planar schematic representation is defined to illustrate the routing of tendons (Tsai and Lee, 1989). To construct a planar schematic, we start from the second joint axis and twist every joint axis about the common normal defined by the joint axis itself and its preceding joint axis until all the axes are pointing in the same direction. In this way, the routing of tendons can be shown clearly without losing the tendon routing topology. Figure 8.6a shows a pla-

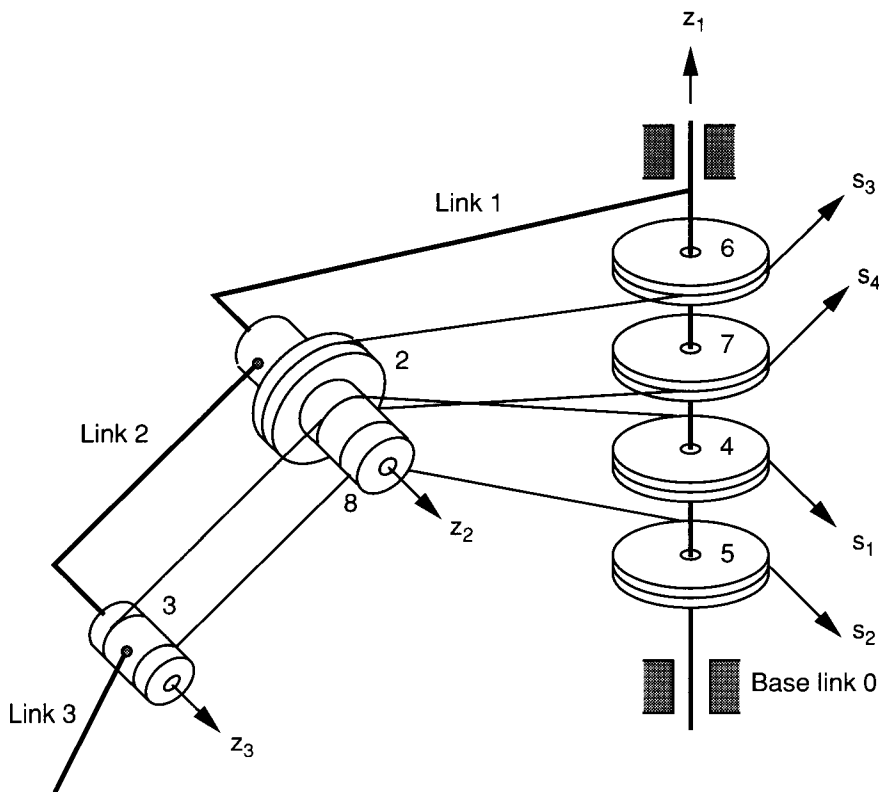


FIGURE 8.5. Functional schematic diagram of the Stanford/JPL finger.

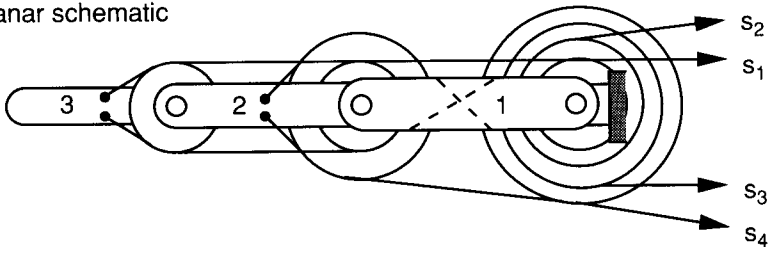
nar schematic diagram of the Stanford/JPL finger. In a planar representation, all the positive joint axes point out of the paper. The routing of each tendon can also be sketched individually, as shown in Fig. 8.6*b–e*.

A more abstract planar representation of the tendon routing is shown in Fig. 8.7. In this abbreviated planar representation, a line passing by one side of a pulley implies that the tendon is routed over that pulley several times, beginning and ending at that side of the pulley. Similar to a gear train, we call each tendon routing arrangement a *transmission line*.

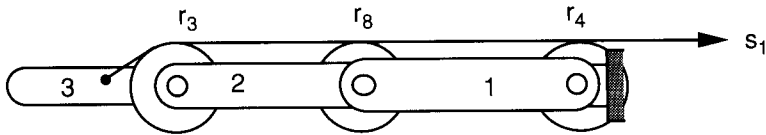
8.4 KINEMATICS OF TENDON-DRIVEN MANIPULATORS

The kinematics of tendon-driven manipulators is similar to that of geared robotic mechanisms. The analysis can be accomplished in two fundamental steps. The first step is the derivation of a kinematic relationship between

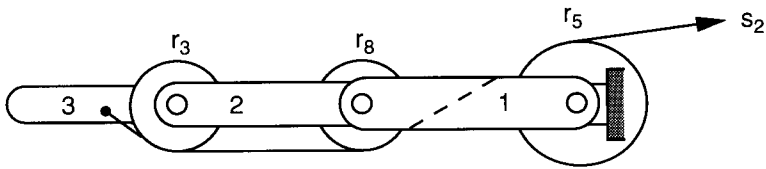
(a) Planar schematic



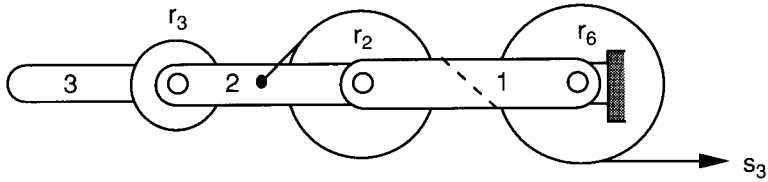
(b) Transmission line 1



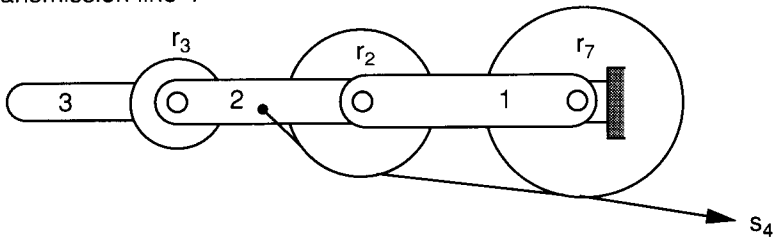
(c) Transmission line 2



(d) Transmission line 3



(e) Transmission line 4

**FIGURE 8.6.** Planar schematic diagram of the Stanford/JPL finger.

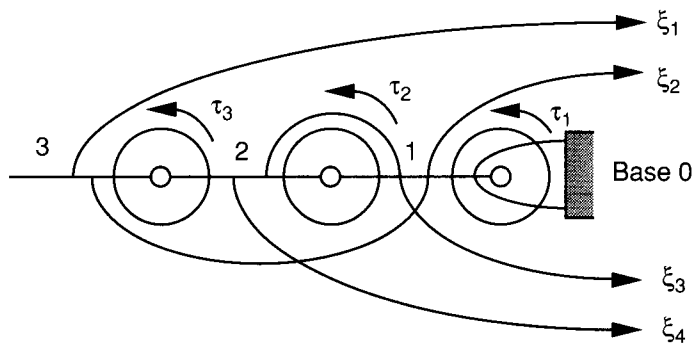


FIGURE 8.7. Abbreviated planar representation of the Stanford/JPL finger.

the location of the end effector and the joint angles of the open-loop chain. This step can be accomplished by the Denavit–Hartenberg method outlined in Chapter 2. The second step is the derivation of a kinematic relationship between the joint angles and the tendon displacements, and this is the main focus of this section.

Mathematically, the joint angles and tendon displacements are related by a linear transformation. In this section we apply the theory of fundamental circuits and the coaxiality conditions for the derivation of the transformation matrix. We will show that the transformation matrix can be obtained by an inspection of the tendon routing topology using the concept of transmission lines. In general, the transformation matrix is a function of the pulley size, the tendon routing topology, and the manipulator posture. However, for those open-loop manipulators in which tendons are routed from one joint to another along the open-loop chain in a consecutive manner, we show that the matrix is independent of the manipulator posture.

8.4.1 Fundamental Circuit Equations

Let i and j denote a pulley pair for which link k serves as the common carrier, as shown in Fig. 8.8. Then links i , j , and k constitute a simple tendon-and-pulley train. For the purpose of analysis, a positive direction of rotation is assigned to each joint axis of the pulley pair. A *fundamental circuit equation* relating the angular displacements of the two pulleys can be written as

$$r_i \theta_{i,k} = \pm r_j \theta_{j,k}, \quad (8.1)$$

where $\theta_{i,k}$ and $\theta_{j,k}$ denote the rotations of pulleys i and j with respect to the carrier k and r_i and r_j denote the radii of the two pulleys.

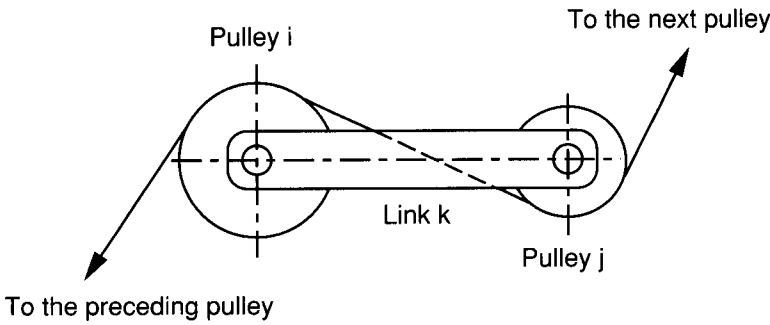


FIGURE 8.8. Planar schematic of a simple tendon-and-pulley train.

A tendon routing is called a *parallel type* if a rotation of one pulley about its prescribed positive joint axis produces a positive rotation of the other pulley. Otherwise, it is called a *crossed type*. For example, the routing shown in Fig. 8.8 is a crossed type. The sign in Eq. (8.1) is positive for parallel routing and negative for crossed routing. Equation (8.1) is valid whether or not the carrier is stationary.

8.4.2 Coaxiality Conditions

Let i , j , and k denote three links that share a common joint axis; then similar to an epicyclic gear train, the angular displacements of the three links are related by the following chain rule:

$$\theta_{i,j} = \theta_{i,k} - \theta_{j,k}. \quad (8.2)$$

Equation (8.2) is useful for relating the relative rotations among three coaxial links.

8.4.3 Transmission Lines

Figure 8.9 shows a typical transmission line in which the links are labeled sequentially from 0 to n and the pulleys are labeled from j to $j + n - 1$. Let $\theta_{j,i}$ denote the angular displacement of link j with respect to link i . We can write the fundamental circuit equation once for each pulley pair as follows:

$$r_{j+i-1}\theta_{j+i-1,i} = \pm r_{j+i}\theta_{j+i,i} \quad \text{for } i = 1, 2, \dots, n-1. \quad (8.3)$$

where the $\pm$ sign is positive for a parallel routing and negative for a crossed routing.

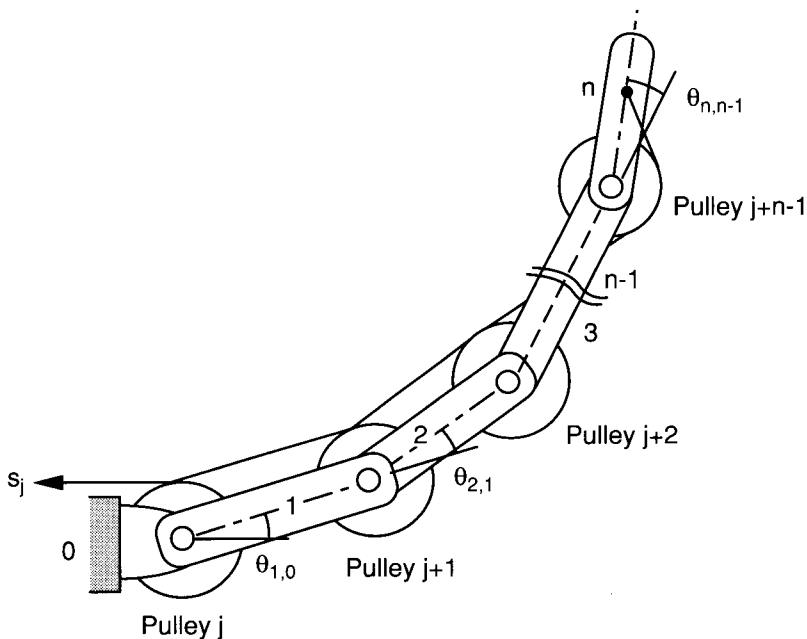


FIGURE 8.9. Typical transmission line.

Since links $i-1$, i , and pulley $j+i-1$ are coaxial, a coaxiality condition can be written as

$$\theta_{j+i-1,i} = \theta_{j+i-1,i-1} - \theta_{i,i-1} \quad \text{for } i = 1, 2, \dots, n. \quad (8.4)$$

Since pulley $j+n-1$ is attached to link n , $\theta_{j+n-1,n} = 0$. Eliminating $\theta_{j+i-1,i}$ for $i = 2, 3, \dots, n$ from Eqs. (8.3) and (8.4) yields

$$\theta_{j,0} = \theta_{1,0} \pm (r_{j+1}/r_j)\theta_{2,1} \pm \dots \pm (r_{j+n-1}/r_j)\theta_{n,n-1}. \quad (8.5)$$

The $\pm$ sign for each term of $\theta_{k,k-1}$ in Eq. (8.5) is determined by the number of crossed routings preceding the k th joint axis. The sign is positive if the number of crossed routings is even; otherwise, it is negative. Hence the equation can readily be obtained by an inspection of the tendon routing topology.

Let the displacement of tendon j from a fully stretched out reference position be denoted by s_j . It is obvious that the displacement of tendon j is related to the angular displacement of the base pulley by

$$s_j = \pm r_j \theta_{j,0}, \quad (8.6)$$

where the $\pm$ sign is positive if a pull of the tendon results in a positive rotation of the base pulley, otherwise, it is negative.

Eliminating $\theta_{j,0}$ between Eqs. (8.6) and (8.5) and writing the resulting equation once for each tendon yields m linear equations relating the tendon displacements to the joint angles of the open-loop chain. These equations can be assembled in matrix form:

$$\mathbf{s} = A\boldsymbol{\theta}, \quad (8.7)$$

where $\mathbf{s} = [s_1, s_2, \dots, s_m]^T$ denotes an m -dimensional vector of tendon displacements, $\boldsymbol{\theta} = [\theta_{1,0}, \theta_{2,1}, \dots, \theta_{n,n-1}]^T$ denotes an n -dimensional vector of the joint angles, and A is an $m \times n$ transformation matrix. The elements of A are functions of the pulley sizes and the tendon routing topology and are independent of the posture of a manipulator. Since $m \geq n + 1$, A is not a square matrix.

Equation (8.7) constitutes a set of m linear equations. Given the joint angles of the open-loop chain, we can compute the tendon displacements directly from Eq. (8.7). On the other hand, since $m > n$, the tendon displacements cannot be specified arbitrarily. Specifically, only n of the m tendon displacements are needed to determine the joint angles. The displacements of the remaining $m - n$ tendons follow the constraints imposed by Eq. (8.7). This contributes to a major difference between a tendon-driven manipulator and a geared robotic mechanism.

Example 8.4.1 *Kinematics of the Stanford/JPL Finger* Let us apply the concept of transmission lines to the kinematic analysis of the Stanford/JPL finger shown in Fig. 8.5. The open-loop chain consists of links 0, 1, 2, and 3. The routing of four tendons is depicted clearly in Fig. 8.6.

Writing Eq. (8.5) once for each transmission line shown in Fig. 8.6 yields

$$\begin{aligned} \theta_{4,0} &= \theta_{1,0} + (r_8/r_4)\theta_{2,1} + (r_3/r_4)\theta_{3,2}, \\ \theta_{5,0} &= \theta_{1,0} - (r_8/r_5)\theta_{2,1} - (r_3/r_5)\theta_{3,2}, \\ \theta_{6,0} &= \theta_{1,0} - (r_2/r_6)\theta_{2,1}, \\ \theta_{7,0} &= \theta_{1,0} + (r_2/r_7)\theta_{2,1}. \end{aligned} \quad (8.8)$$

Note that we have made use of the following facts in the derivation of Eq. (8.8). The parallel-and-parallel routing of tendon 1 leads to positive second and third terms on the right-hand side of the first equation; the crossed-and-parallel routing of tendon 2 results in negative second and third terms on the right-hand side of the second equation; the crossed routing of tendon 3 leads to a negative second term on the right-hand side of the third equation;

and the parallel routing of tendon 4 results in a positive second term on the right-hand side of the fourth equation.

The tendon displacements are related to the base pulley rotations by the following equations:

$$\begin{aligned} s_1 &= -r_4\theta_{4,0}, \\ s_2 &= -r_5\theta_{5,0}, \\ s_3 &= r_6\theta_{6,0}, \\ s_4 &= r_7\theta_{7,0}. \end{aligned} \quad (8.9)$$

Eliminating $\theta_{4,0}$, $\theta_{5,0}$, $\theta_{6,0}$, and $\theta_{7,0}$ between Eqs. (8.8) and (8.9), we obtain

$$\mathbf{s} = A\boldsymbol{\theta}, \quad (8.10)$$

where $\mathbf{s} = [s_1, s_2, s_3, s_4]^T$, $\boldsymbol{\theta} = [\theta_{1,0}, \theta_{2,1}, \theta_{3,2}]^T$, and

$$A = \begin{bmatrix} -r_4 & -r_8 & -r_3 \\ -r_5 & r_8 & r_3 \\ r_6 & -r_2 & 0 \\ r_7 & r_2 & 0 \end{bmatrix}. \quad (8.11)$$

We observe that the elements of A are functions of tendon routing and pulley sizes. The tendon displacements are highly coupled. The third joint angle, $\theta_{3,2}$, affects the displacements of tendons 1 and 2 only, while the first and second joint angles, $\theta_{1,0}$ and $\theta_{2,1}$, affect the displacements of all four tendons.

Example 8.4.2 Kinematics of the Utah/MIT Finger As a second example we consider a 4-dof manipulator controlled by eight tendons, resembling the Utah/MIT finger. In the Utah/MIT finger, the first joint axis is fixed to the base; the second joint axis is perpendicular to the first with a small offset distance; while the third and fourth joint axes are both parallel to the second joint axis. A planar schematic diagram of the mechanism is shown in Fig. 8.10. As shown in the figure, pulleys 1, 2, 3, and 4 are attached to links 1, 2, 3, and 4, respectively. Pulleys 1, 5, 6, and 7 are pivoted about the first joint axis; pulleys 2, 8, and 9 are pivoted about the second joint axis; pulleys 3 and 10 are pivoted about the third joint axis; and pulley 4 is pivoted about the fourth joint axis. The first and second tendons, s_1 and s_2 , wrap around the opposite sides of pulleys 5, 8, 10, and 4; the third and fourth tendons, s_3 and s_4 , wrap around the opposite sides of pulleys 6, 9, and 3; the fifth and sixth tendons, s_5 and s_6 , wrap around the opposite sides of pulleys 7 and 2; and the seventh and eighth tendons, s_7 and s_8 , wrap around the opposite sides of pulley 1.

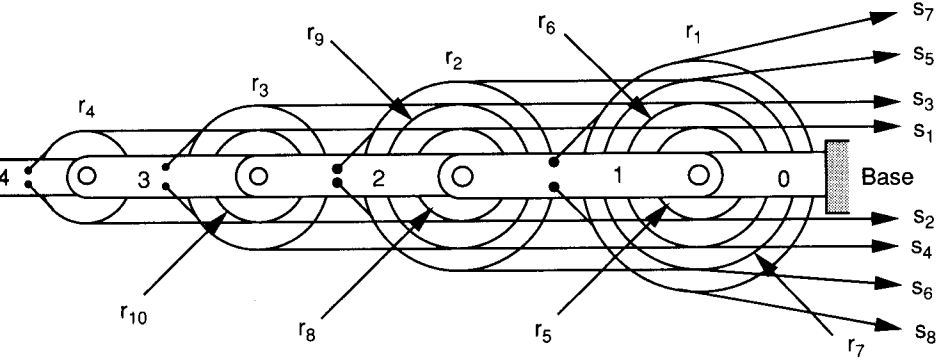


FIGURE 8.10. A 4-dof manipulator driven by eight tendons.

We wish to find the transformation between the joint angles and the tendon displacements.

Writing Eq. (8.5) once for each of the eight tendons yields

$$\begin{aligned}
 \theta_{5,0} &= \theta_{1,0} + (r_8/r_5)\theta_{2,1} + (r_{10}/r_5)\theta_{3,2} + (r_4/r_5)\theta_{4,3}, \\
 \theta_{5,0} &= \theta_{1,0} + (r_8/r_5)\theta_{2,1} + (r_{10}/r_5)\theta_{3,2} + (r_4/r_5)\theta_{4,3}, \\
 \theta_{6,0} &= \theta_{1,0} + (r_9/r_6)\theta_{2,1} + (r_3/r_6)\theta_{3,2}, \\
 \theta_{6,0} &= \theta_{1,0} + (r_9/r_6)\theta_{2,1} + (r_3/r_6)\theta_{3,2}, \\
 \theta_{7,0} &= \theta_{1,0} + (r_2/r_7)\theta_{2,1}, \\
 \theta_{7,0} &= \theta_{1,0} + (r_2/r_7)\theta_{2,1}, \\
 \theta_{1,0} &= \theta_{1,0}, \\
 \theta_{1,0} &= \theta_{1,0}.
 \end{aligned} \tag{8.12}$$

Note that the first and second equations in Eq. (8.12) are identical, due to the fact that tendons 1 and 2 are wrapped around opposite sides of the same pulleys. Similar conditions occur for the remaining tendons. The tendon displacements are related to the angular displacements of the base pulleys by

$$\begin{aligned}
 s_1 &= -r_5\theta_{5,0}, \\
 s_2 &= r_5\theta_{5,0}, \\
 s_3 &= -r_6\theta_{6,0}, \\
 s_4 &= r_6\theta_{6,0}, \\
 s_5 &= -r_7\theta_{7,0},
 \end{aligned}$$

$$\begin{aligned}
s_6 &= r_7 \theta_{7,0}, \\
s_7 &= -r_1 \theta_{1,0}, \\
s_8 &= r_1 \theta_{1,0}.
\end{aligned} \tag{8.13}$$

Substituting Eq. (8.12) into (8.13), we obtain

$$\mathbf{s} = A\boldsymbol{\theta}, \tag{8.14}$$

where $\mathbf{s} = [s_1, s_2, \dots, s_8]^T$, $\boldsymbol{\theta} = [\theta_{1,0}, \theta_{2,1}, \theta_{3,2}, \theta_{4,3}]^T$, and

$$A = \begin{bmatrix} -r_5 & -r_8 & -r_{10} & -r_4 \\ r_5 & r_8 & r_{10} & r_4 \\ -r_6 & -r_9 & -r_3 & 0 \\ r_6 & r_9 & r_3 & 0 \\ -r_7 & -r_2 & 0 & 0 \\ r_7 & r_2 & 0 & 0 \\ -r_1 & 0 & 0 & 0 \\ r_1 & 0 & 0 & 0 \end{bmatrix}. \tag{8.15}$$

Equation (8.14) provides a transformation between the joint angles and the tendon displacements. Given the joint angles, we can compute the corresponding tendon displacements from Eq. (8.14).

8.5 STATIC FORCE ANALYSIS

In this section we apply the principle of virtual work to derive a transformation between the equivalent joint torques and the tendon forces. Let the vector of tendon forces be denoted by $\boldsymbol{\xi} = [\xi_1, \xi_2, \dots, \xi_m]^T$ and the vector of output joint torques be denoted by $\boldsymbol{\tau} = [\tau_1, \tau_2, \dots, \tau_n]^T$. Also let the virtual joint displacements be denoted by $\delta\boldsymbol{\theta} = [\delta\theta_{1,0}, \delta\theta_{2,1}, \dots, \delta\theta_{n,n-1}]^T$ and the virtual tendon displacements be denoted by $\delta\mathbf{s} = [s_1, s_2, \dots, s_m]^T$. Then the virtual work, δW , done by the active forces is given by

$$\delta W = -\boldsymbol{\tau}^T \delta\boldsymbol{\theta} + \boldsymbol{\xi}^T \delta\mathbf{s}. \tag{8.16}$$

However, the virtual displacements $\delta\boldsymbol{\theta}$ and $\delta\mathbf{s}$ in Eq. (8.16) are related by

$$\delta\mathbf{s} = A\delta\boldsymbol{\theta}. \tag{8.17}$$

Substituting Eq. (8.17) into (8.16) yields

$$\delta W = (-\boldsymbol{\tau}^T + \boldsymbol{\xi}^T A) \delta\boldsymbol{\theta}. \tag{8.18}$$

Since the system is under equilibrium, the virtual work vanishes for any arbitrary virtual displacement $\delta\theta$. Hence we conclude that

$$-\tau^T + \xi^T A = 0. \quad (8.19)$$

Taking the transpose of Eq. (8.19), we obtain

$$\tau = B\xi. \quad (8.20)$$

where $B = A^T$ is called the *structure matrix*. We note that each column of B represents a transmission line.

Equation (8.20) provides a transformation from the tendon forces to the joint torques. However, as described in Chapter 6, the joint torques are related to the end-effector output forces by the transpose of the Jacobian matrix:

$$\tau = J^T F. \quad (8.21)$$

Eliminating τ from Eqs. (8.20) and (8.21), we obtain the overall transformation

$$B\xi = J^T F. \quad (8.22)$$

Assuming that J is a nonsingular square matrix, Eq. (8.22) can be inverted as follows:

$$F = J^{-T} B\xi. \quad (8.23)$$

Therefore, given the tendon forces, we can compute the end-effector output forces directly from Eq. (8.23). Since $m > n$, given the end-effector output force, there exist an infinitude of tendon forces. But only those positive tendon forces are permissible. This makes control of tendon-drive manipulators more complicated than that of conventional manipulators. A systematic method for the resolution of tendon forces is discussed in a later section.

Furthermore, assuming that pulleys mounted on the same joint axis are of the same size, the structure matrix can be decomposed into a product of two matrices,

$$B = R^* B^*, \quad (8.24)$$

where R^* is an $n \times n$ diagonal matrix whose nonzero elements are the radii of the pulleys installed on the consecutive joint axes of the open-loop chain and B^* is an $n \times m$ matrix whose (i, j) element takes the value $+1$, -1 , or 0 depending on whether a positive pull of tendon j produces a positive, negative, or no effect on the i th joint torque. Thus B^* contains the tendon routing topology, while R^* represents the pulley sizes.

Example 8.5.1 *Structural Matrix of the Stanford/JPL Finger* The structure matrix of the Stanford/JPL finger shown in Fig. 8.5 is obtained by taking the transpose of Eq. (8.11):

$$B = A^T = \begin{bmatrix} -r_4 & -r_5 & r_6 & r_7 \\ -r_8 & r_8 & -r_2 & r_2 \\ -r_3 & r_3 & 0 & 0 \end{bmatrix}. \quad (8.25)$$

Furthermore, if $r_2 = r_8$ and $r_4 = r_5 = r_6 = r_7$, the structure matrix can be written as $B = R^* B^*$, where

$$B^* = \begin{bmatrix} -1 & -1 & 1 & 1 \\ -1 & 1 & -1 & 1 \\ -1 & 1 & 0 & 0 \end{bmatrix}, \quad (8.26)$$

$$R^* = \begin{bmatrix} r_4 & 0 & 0 \\ 0 & r_2 & 0 \\ 0 & 0 & r_3 \end{bmatrix}. \quad (8.27)$$

The first column of B^* implies that a pull of tendon 1 will produce negative torques about the first, second, and third joint axes. The second column of B^* implies that a pull of tendon 2 will result in negative, positive, and positive torques about the first, second, and third joint axes, respectively. The effects of tendons 3 and 4 on the joint torques of the manipulator can be explained similarly.

Example 8.5.2 *Structural Matrix of the Utah/MIT Finger* The structure matrix of the 4-dof manipulator shown in Fig. 8.10 is obtained by taking the transpose of Eq. (8.15):

$$B = \begin{bmatrix} -r_5 & r_5 & -r_6 & r_6 & -r_7 & r_7 & -r_1 & r_1 \\ -r_8 & r_8 & -r_9 & r_9 & -r_2 & r_2 & 0 & 0 \\ -r_{10} & r_{10} & -r_3 & r_3 & 0 & 0 & 0 & 0 \\ -r_4 & r_4 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (8.28)$$

Furthermore, if $r_1 = r_5 = r_6 = r_7$, $r_2 = r_8 = r_9$, and $r_3 = r_{10}$, the structure matrix can be written as $B = R^* B^*$, where

$$B^* = \begin{bmatrix} -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 \\ -1 & 1 & -1 & 1 & -1 & 1 & 0 & 0 \\ -1 & 1 & -1 & 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (8.29)$$

$$R^* = \begin{bmatrix} r_1 & 0 & 0 & 0 \\ 0 & r_2 & 0 & 0 \\ 0 & 0 & r_3 & 0 \\ 0 & 0 & 0 & r_4 \end{bmatrix}. \quad (8.30)$$

The matrix B^* contains eight columns formed by four pairs of tendons: (1,2), (3,4), (5,6), and (7,8). For each pair of tendons, the corresponding column vectors are negatives of each other, indicating that the tendons are routed symmetrically around opposite sides of the pulleys.

8.6 FEASIBLE STRUCTURE MATRICES

In this section we discuss the fundamental structure characteristics of tendon-driven manipulators. We show that an n -dof open-loop manipulator requires at least $n + 1$ tendons, and these tendons must be routed in certain patterns such that independent control of the joints can be achieved. Finally, we introduce the notion of isomorphic structure matrices and present some feasible structure matrices.

8.6.1 Structure Characteristics

The structure matrix defines the force transmission characteristics between the tendon force space and the joint torque space. Given a set of tendon forces, the resulting joint torques of the open-loop chain can be computed directly from Eq. (8.20). Conversely, given a set of desired joint torques, Eq. (8.20) constitutes n linear equations in m unknowns. To maintain positive tendon forces, m must be greater than n such that the solutions of tendon forces can be written as a particular solution plus an $(m - n)$ -dimensional homogeneous solution. More specifically, for $m > n$, we can express the inverse transformation of Eq. (8.20) as

$$\xi = B^+ \tau + H\lambda, \quad (8.31)$$

where $B^+ = B^T [BB^T]^{-1}$ is the pseudoinverse of B , H is an $m \times (m - n)$ matrix with its column vectors spanning the null space of B , and λ is an arbitrary $(m - n)$ -dimensional vector.

The first term on the right-hand side of Eq. (8.31) is called the *particular solution*, and the second term, the *homogeneous solution*. The particular solutions form an n -dimensional subspace in the m -dimensional tendon-force space. The homogeneous solution gives the ratios of tendon forces that result in no net joint torques. Hence if the column space of H contains at least one

m -dimensional vector with all positive elements, by choosing a proper value of λ , positive tendon forces can always be maintained.

Based on the discussion above, we conclude that the structure matrix B must satisfy the following conditions (Lee and Tsai, 1991; Ou and Tsai, 1993):

1. The number of tendons must exceed the number of degrees of freedom by at least 1 (i.e., $m \geq n + 1$).
2. The rank of B must be equal to the degrees of freedom of the system. Furthermore, if $m = n + 1$, the determinant of a submatrix formed by deleting any column from B should not be equal to zero.
3. There exists at least one vector with all positive elements in the null space of B .
4. Since we assume that tendons are routed from joint to joint in a consecutive manner, nonzero elements in each column of B must be consecutive.
5. Exchanging any two columns of a structure matrix, which is equivalent to a renumbering of two tendons, does not have any effect on the mechanics of a manipulator. It follows that the structure matrix can always be arranged in a form such that all the zero elements appear on the lower right corner of the matrix.
6. Changing the sign of every element in a row does not have any effect on the mechanics of a manipulator. This is equivalent to a change in the definition of the positive direction of rotation of a joint axis.

8.6.2 Isomorphic Structure Matrices

We have shown in Eq. (8.5) that the elements of a structure matrix are determined by the routing of tendons and the definitions of positive directions of rotation of the joint axes. Therefore, if the positive direction of rotation of a joint axis is defined in the opposite sense, the sign for each element in the corresponding row of the structure matrix will be reversed. Since the definition of positive direction of rotation and the numbering of tendons have no effect on the functionality of a mechanism, two structure matrices are said to be structurally *isomorphic* if they satisfy one of the following conditions:

1. They are identical.
2. They become identical after changing the sign of every element in one or more rows of a matrix (structure characteristic 6).

3. They become identical after exchanging two or more columns of a matrix (structure characteristic 5).
4. They become identical after the combined operation of conditions 2 and 3.

Example 8.6.1 *Two Isomorphic Tendon Routings* Figure 8.11a and b show a 2-dof manipulator with two apparently different routings of tendons. We wish to show that these two routings are structurally isomorphic. Let the positive direction of rotation be pointing out of the paper. Then the structure matrices for the two routings of tendons are given by

$$(1) \quad B_1^* = \begin{bmatrix} -1 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix},$$

$$(2) \quad B_2^* = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 0 \end{bmatrix},$$

respectively. It can be seen that after switching the sign of each element in the first row of B_2^* , the structure matrix B_2^* becomes identical to B_1^* . Physically, if we reverse the definition of positive direction of rotation of the first (proximal) joint axis of the mechanism shown in Fig. 8.11b, it becomes identical to that of Fig. 8.11a. Therefore, these two routings are structurally isomorphic.

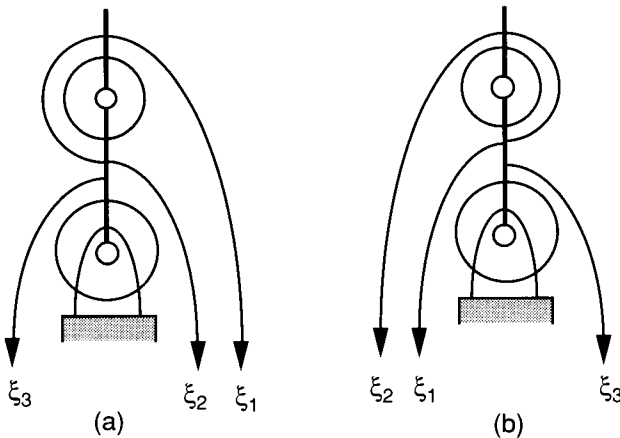


FIGURE 8.11. Two structurally isomorphic, 2-dof tendon-driven manipulators.

8.6.3 Feasible Pseudotriangular Structure Matrices

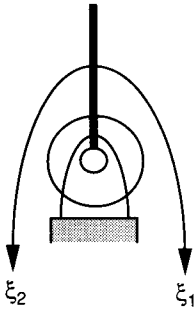
In the preceding section we have shown that an n -dof manipulator requires at least $n + 1$ tendons and that the routing of these tendons must satisfy the structure characteristics 1 through 6. We call a structure matrix that satisfies characteristics 1 through 6 a *feasible structure matrix*; otherwise, the manipulator is not controllable.

Assuming that all pulleys are of the same size, an atlas of nonisomorphic pseudotriangular structure matrices representing tendon-driven manipulators with up to 6 degrees of freedom was developed by Lee and Tsai (1991). Figure 8.12 shows the planar schematics of nonisomorphic tendon-driven manipulators with up to 3 degrees of freedom. Figures 8.13 and 8.14 list all feasible nonisomorphic routings of tendon-driven manipulators with 4 degrees of freedom. We note that the Stanford/JPL finger and Utah/MIT finger do not belong to this class of manipulators. Note that the number of feasible tendon routings increases tremendously if pulleys are permitted to take different sizes (Ou and Tsai, 1993).

8.7 REDUNDANT FORCES RESOLUTION

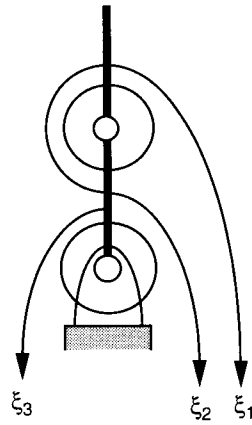
In this section we develop a systematic method for the resolution of redundant tendon forces. Since the joint space and the tendon space do not have the same dimension, the mapping from the joint torques to the tendon forces is not unique. Theoretically, the pseudoinverse technique can be used. However, the computation of the pseudoinverse can be very time consuming. In addition, the vector λ in Eq. (8.31) must be chosen properly such that all tendons are under tension at all times. To achieve this, the largest ratio of all the negative tendon forces in the particular solution to their corresponding components in the homogeneous solution must be identified. This process will inevitably increase the computation time and reduce the possibility for real-time control of a tendon-driven manipulator system. To overcome this difficulty, Salisbury (1982) supplemented Eq. (8.20) with an additional equation derived from one of the bearing forces for the resolution of redundant forces. Jacobsen et al. (1984, 1989) developed a *rectifier* concept for control of the Utah/MIT hand.

The rectifier concept uses circuitlike operators to convert joint torque signals into tendon force signals without using the pseudoinverse transformation. It provides a closed-form-like solution for the tendon forces. In this section we describe the rectifier concept for the resolution of redundant forces (Lee, 1991; Lee and Tsai, 1993). First, we study the characteristics of a 1-dof system to illustrate the basic principle of a rectifier. Then we extend it to multi-dof systems.



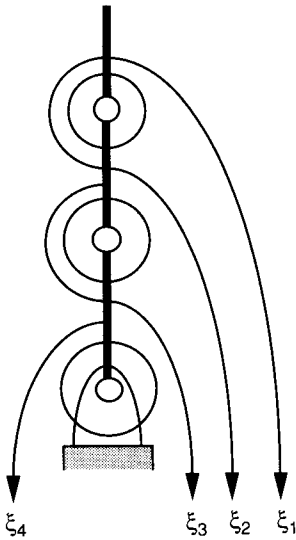
$$B^* = [-1 \ 1]$$

(a) One-dof system

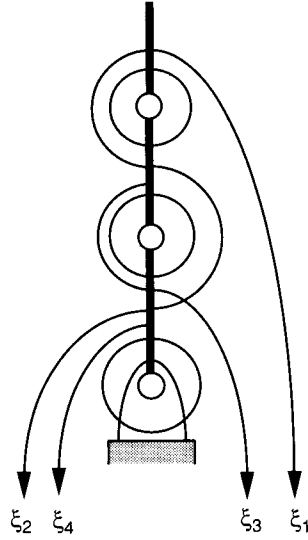


$$B^* = \begin{bmatrix} -1 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

(b) Two-dof system



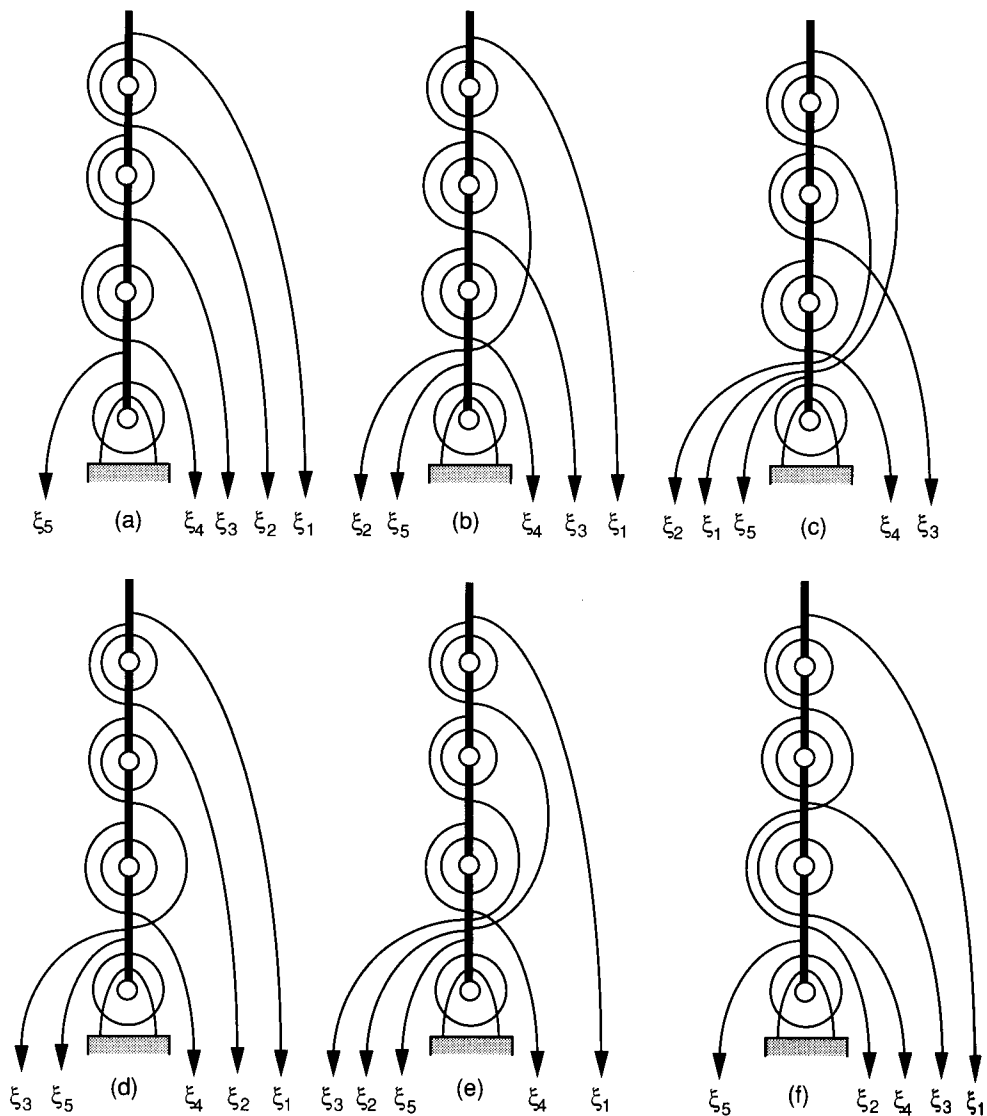
$$B^* = \begin{bmatrix} -1 & -1 & -1 & 1 \\ -1 & -1 & 1 & 0 \\ -1 & 1 & 0 & 0 \end{bmatrix}$$



$$B^* = \begin{bmatrix} -1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 0 \\ -1 & 1 & 0 & 0 \end{bmatrix}$$

(c) Three-dof systems

FIGURE 8.12. Feasible $n \times (n + 1)$ manipulators with up to 3 degrees of freedom.

FIGURE 8.13. Feasible 4×5 tendon routings—part I.

8.7.1 Basic Principle

We first study a 1-dof system with two control tendons, shown in Fig. 8.15. Let the positive direction of rotation be pointing out of the paper and let τ be the resulting joint torque about the fixed pivot, O . Assuming that the pulley has a radius of r , a moment balance equation about the fixed pivot can be

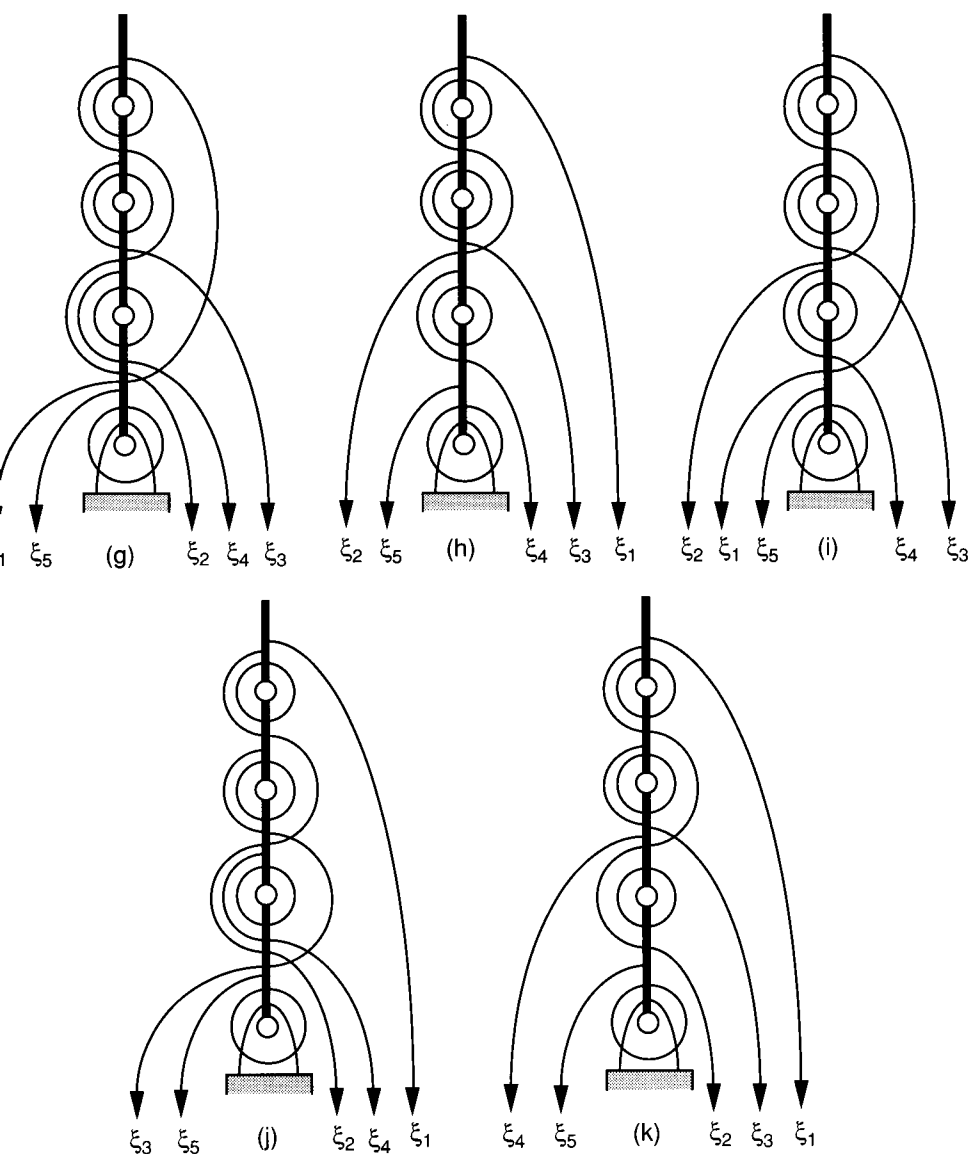


FIGURE 8.14. Feasible 4×5 tendon routings—part II.

written

$$\xi_1 - \xi_2 = \tau/r. \quad (8.32)$$

Equation (8.32) contains two tendon forces, ξ_1 and ξ_2 , both of which must be positive at all times. If τ is positive (counterclockwise), the minimum ten-

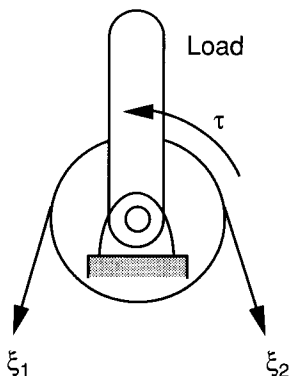


FIGURE 8.15. A 1-dof system controlled by two tendons.

don forces would be $\xi_1 = \tau/r$ and $\xi_2 = 0$. On the other hand, if τ is negative (clockwise), the minimum forces would be $\xi_1 = 0$ and $\xi_2 = -\tau/r$. This simple relation can be written in a mathematical form as

$$\text{for } \tau \geq 0, \begin{cases} \xi_1 = \tau/r + \delta, \\ \xi_2 = \delta, \end{cases} \quad (8.33)$$

$$\text{for } \tau < 0, \begin{cases} \xi_1 = \delta, \\ \xi_2 = -\tau/r + \delta, \end{cases} \quad (8.34)$$

where δ is a positive bias force that has no net effect on the resultant joint torque.

For convenience, we define two operators O^+ and O^- :

$$O^+(x) = \begin{cases} x, & x \geq 0, \\ 0, & x < 0, \end{cases} \quad (8.35)$$

$$O^-(x) = \begin{cases} 0, & x \geq 0, \\ -x, & x < 0, \end{cases} \quad (8.36)$$

where x is a dummy variable.

The characteristics of $O^+(x)$ and $O^-(x)$ can be described graphically as shown in Fig. 8.16. Mathematically, $O^+(x)$ and $O^-(x)$ can also be written as

$$O^+(x) = \frac{x + |x|}{2}, \quad (8.37)$$

$$O^-(x) = \frac{-x + |x|}{2}. \quad (8.38)$$

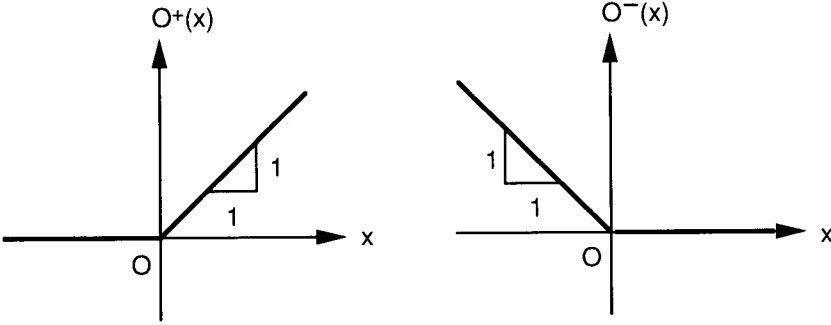


FIGURE 8.16. Characteristics of $O^+(x)$ and $O^-(x)$ operators.

Following the definitions above, it can be shown that

$$O^+(x) + O^-(x) = |x|, \quad (8.39)$$

$$O^+(x) - O^-(x) = x. \quad (8.40)$$

Expressing Eqs. (8.33) and (8.34) in terms of O^+ and O^- operators, we obtain

$$\begin{aligned} \xi_1 &= O^+(\tau/r) + \delta, \\ \xi_2 &= O^-(\tau/r) + \delta. \end{aligned} \quad (8.41)$$

We note that the O^+ operator goes with the tendon force that produces a positive joint torque, and the O^- operator goes with the one that results in a negative joint torque in Eq. (8.32). The physical meaning of Eq. (8.41) can readily be seen from Fig. 8.15. If the required joint torque is positive, ξ_1 must give a minimum pull of τ/r while ξ_2 remains zero. On the other hand, if the required joint torque is negative, ξ_2 must give a minimum pull of $|\tau/r|$ while ξ_1 remains zero. Adding a bias force δ to both ξ_1 and ξ_2 has no effect on the resulting joint torque. Furthermore, the ratio of the two bias forces, 1:1 for this simple system, is proportional to the homogeneous solution of Eq. (8.32). The addition of a bias force prevents the tendons from slackening.

The rectifier concept can be implemented in a conventional or other advanced controller. Figure 8.17 illustrates a PD controller using the rectifier concept for the control of the 1-dof manipulator shown in Fig. 8.15.

8.7.2 Application to Multi-DOF Systems

We now demonstrate how the tendon forces in an n -dof system can be determined systematically using the $O^+(x)$ and $O^-(x)$ operators. In general,

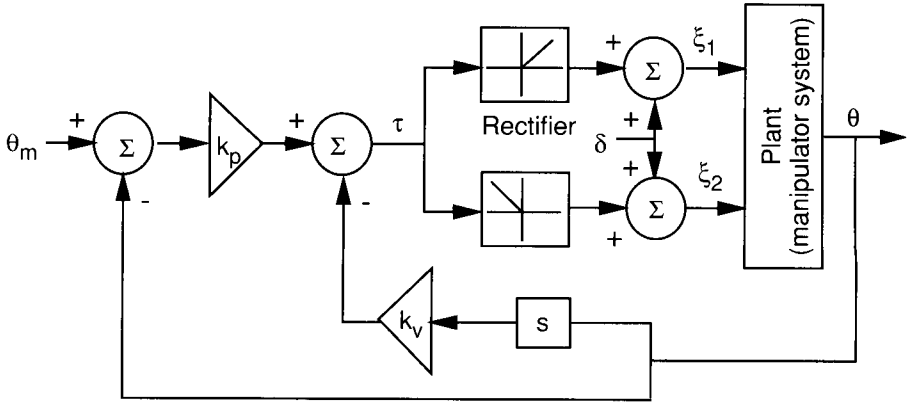


FIGURE 8.17. PD controller using the rectifier concept.

we first reduce the system of equations given by Eq. (8.20) to one equation in two unknown forces. Next, we solve these two unknown forces by applying the $O^+(x)$ and $O^-(x)$ operators. Then we substitute the results back into the original system of equations to obtain a reduced system of equations and repeat the process until all the variables are solved. The following examples serve to illustrate the methodology. For simplicity, we assume that pulleys mounted on the same joint axis are of the same size.

Example 8.7.1 *A 3-DOF Manipulator with a Pseudotriangular Structure Matrix* We first consider a spatial 3-dof manipulator controlled by four tendons, as shown in Fig. 8.18. The structure matrix B , obtained by an inspection of the tendon routing, is

$$B = R^* B^* = \begin{bmatrix} r_1 & 0 & 0 \\ 0 & r_2 & 0 \\ 0 & 0 & r_3 \end{bmatrix} \begin{bmatrix} -1 & -1 & -1 & 1 \\ -1 & -1 & 1 & 0 \\ -1 & 1 & 0 & 0 \end{bmatrix}. \quad (8.42)$$

Because of the routing topology, the structure matrix takes the pseudotriangular form. The null space of B^* is given by $H = [1, 1, 2, 4]^T$. Substituting the structure matrix above into Eq. (8.20) and multiplying both sides of the resulting equation by the inverse of R^* yields

$$-\xi_1 - \xi_2 - \xi_3 + \xi_4 = \tau_1/r_1, \quad (8.43)$$

$$-\xi_1 - \xi_2 + \xi_3 = \tau_2/r_2, \quad (8.44)$$

$$-\xi_1 + \xi_2 = \tau_3/r_3. \quad (8.45)$$

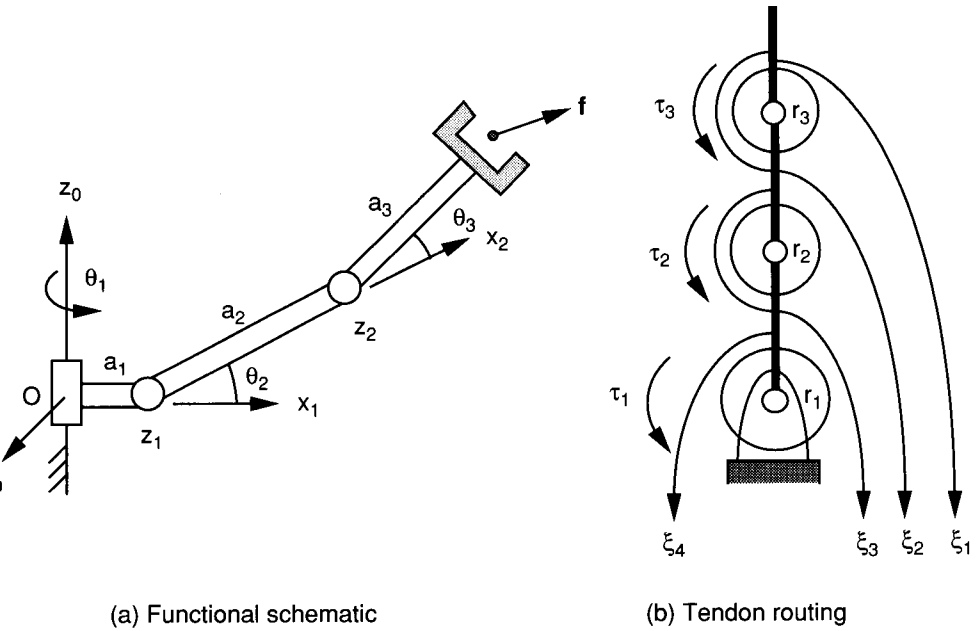


FIGURE 8.18. A 3-dof manipulator controlled by four tendons.

Since Eq. (8.45) contains only two unknown forces, ξ_1 and ξ_2 , and they must be positive at all times, we can express ξ_1 and ξ_2 in terms of the O^+ and O^- operators by applying Eq. (8.41):

$$\begin{aligned}\xi_1 &= O^-(\tau_3/r_3) + \delta_1, \\ \xi_2 &= O^+(\tau_3/r_3) + \delta_1,\end{aligned}\tag{8.46}$$

where δ_1 is a positive bias force that results in no net joint torque about the third joint axis.

To determine ξ_3 , we substitute Eq. (8.46) into (8.44) and then apply the relation $O^-(x) + O^+(x) = |x|$. This yields

$$-2\delta_1 + \xi_3 = \tau_2/r_2 + |\tau_3/r_3|.\tag{8.47}$$

We may consider Eq. (8.47) as one equation in two unknowns, δ_1 and ξ_3 . Applying the O^+ and O^- operators, we obtain

$$\begin{aligned}\delta_1 &= \frac{O^-(\tau_2/r_2 + |\tau_3/r_3|)}{2} + \frac{\delta_2}{2}, \\ \xi_3 &= O^+(\tau_2/r_2 + |\tau_3/r_3|) + \delta_2,\end{aligned}\tag{8.48}$$

where δ_2 is a positive bias force that results in no net joint torques about the second and third joint axes.

Combining Eqs. (8.46) and (8.48) yields

$$\begin{aligned}\xi_1 &= O^-(\tau_3/r_3) + \frac{O^-(\tau_2/r_2 + |\tau_3/r_3|)}{2} + \frac{\delta_2}{2}, \\ \xi_2 &= O^+(\tau_3/r_3) + \frac{O^-(\tau_2/r_2 + |\tau_3/r_3|)}{2} + \frac{\delta_2}{2}, \\ \xi_3 &= O^+(\tau_2/r_2 + |\tau_3/r_3|) + \delta_2.\end{aligned}\quad (8.49)$$

Substituting Eqs. (8.49) into Eq. (8.43) yields

$$-2\delta_2 + \xi_4 = \tau_1/r_1 + |\tau_2/r_2 + |\tau_3/r_3|| + |\tau_3/r_3|. \quad (8.50)$$

Equation (8.50) contains two unknowns, δ_2 and ξ_4 . Applying the O^+ and O^- operators again, we obtain

$$\begin{aligned}\delta_2 &= \frac{O^-(\tau_1/r_1 + |\tau_2/r_2 + |\tau_3/r_3|| + |\tau_3/r_3|)}{2} + \frac{\delta_3}{2}, \\ \xi_4 &= O^+(\tau_1/r_1 + |\tau_2/r_2 + |\tau_3/r_3|| + |\tau_3/r_3|) + \delta_3,\end{aligned}\quad (8.51)$$

where δ_3 is a positive bias force that results in no net joint torques about the first, second, and third joint axes.

Finally, combining Eqs. (8.49) and (8.51), we obtain

$$\begin{aligned}\xi_1 &= O^-(\tau_3/r_3) + \frac{O^-(\tau_2/r_2 + |\tau_3/r_3|)}{2} \\ &\quad + \frac{O^-(\tau_1/r_1 + |\tau_2/r_2 + |\tau_3/r_3|| + |\tau_3/r_3|)}{4} + \frac{\delta_3}{4}, \\ \xi_2 &= O^+(\tau_3/r_3) + \frac{O^-(\tau_2/r_2 + |\tau_3/r_3|)}{2} \\ &\quad + \frac{O^-(\tau_1/r_1 + |\tau_2/r_2 + |\tau_3/r_3|| + |\tau_3/r_3|)}{4} + \frac{\delta_3}{4}, \\ \xi_3 &= O^+(\tau_2/r_2 + |\tau_3/r_3|) + \frac{O^-(\tau_1/r_1 + |\tau_2/r_2 + |\tau_3/r_3|| + |\tau_3/r_3|)}{2} + \frac{\delta_3}{2}, \\ \xi_4 &= O^+(\tau_1/r_1 + |\tau_2/r_2 + |\tau_3/r_3|| + |\tau_3/r_3|) + \delta_3.\end{aligned}\quad (8.52)$$

Equation (8.52) provides an effective method for transforming joint torques to tendon forces. The computation is more straightforward than that of the pseudoinverse technique. The result guarantees that each tendon force is at

least equal to one-fourth of the bias force, δ_3 . Although the bias force can be chosen arbitrarily, its contribution to tendon forces is proportional to the vector of the homogeneous solution (i.e., 1 : 1 : 2 : 4).

Example 8.7.2 Stanford/JPL Finger In this example, we consider the Stanford/JPL finger, shown in Fig. 8.5. The abbreviated planar representation is shown in Fig. 8.7. Assuming that the pulleys mounted on the same joint axis are of the same size (i.e., $r_1 = r_4 = r_5 = r_6 = r_7$ and $r_2 = r_8$), the structure matrix can be written as a product of R^* and B^* , where R^* and B^* are given by Eqs. (8.27) and (8.26), respectively. The null space of B^* is given by $H = [1, 1, 1, 1]^T$.

Substituting $B = R^*B^*$ into Eq. (8.20) yields

$$-\xi_1 - \xi_2 + \xi_3 + \xi_4 = \tau_1/r_1, \quad (8.53)$$

$$-\xi_1 + \xi_2 - \xi_3 + \xi_4 = \tau_2/r_2, \quad (8.54)$$

$$-\xi_1 + \xi_2 = \tau_3/r_3. \quad (8.55)$$

Since Eq. (8.55) contains only two unknowns, ξ_1 and ξ_2 can be written in terms of O^+ and O^- operators as

$$\begin{aligned} \xi_1 &= O^-(\tau_3/r_3) + \delta_1, \\ \xi_2 &= O^+(\tau_3/r_3) + \delta_1, \end{aligned} \quad (8.56)$$

where δ_1 is a positive bias force that produces no net joint torque about the third joint axis.

Substituting Eq. (8.55) into (8.54) yields

$$-\xi_3 + \xi_4 = \tau_2/r_2 - \tau_3/r_3. \quad (8.57)$$

Again, Eq. (8.57) contains only two unknowns, ξ_3 and ξ_4 . Hence they can be written in terms of O^+ and O^- operators as

$$\begin{aligned} \xi_3 &= O^-(\tau_2/r_2 - \tau_3/r_3) + \delta_2, \\ \xi_4 &= O^+(\tau_2/r_2 - \tau_3/r_3) + \delta_2, \end{aligned} \quad (8.58)$$

where δ_2 is a positive bias force that produces no net joint torque about the second joint axis.

Substituting Eqs. (8.56) and (8.58) into (8.53) and making use of Eq. (8.39) yields

$$-2\delta_1 + 2\delta_2 = \tau_1/r_1 + |\tau_3/r_3| - |\tau_2/r_2 - \tau_3/r_3|. \quad (8.59)$$

Hence δ_1 and δ_2 can be written in terms of the O^+ and O^- operators as

$$\begin{aligned}\delta_1 &= \frac{O^-(\tau_1/r_1 + |\tau_3/r_3| - |\tau_2/r_2 - \tau_3/r_3|)}{2} + \delta_3, \\ \delta_2 &= \frac{O^+(\tau_1/r_1 + |\tau_3/r_3| - |\tau_2/r_2 - \tau_3/r_3|)}{2} + \delta_3,\end{aligned}\quad (8.60)$$

where δ_3 is a positive bias force that produces no net joint torques about the first, second, and third joint axes. Substituting Eq. (8.60) into (8.56) and (8.58) yields

$$\begin{aligned}\xi_1 &= O^-(\tau_3/r_3) + \frac{O^-(\tau_1/r_1 + |\tau_3/r_3| - |\tau_2/r_2 - \tau_3/r_3|)}{2} + \delta_3, \\ \xi_2 &= O^+(\tau_3/r_3) + \frac{O^-(\tau_1/r_1 + |\tau_3/r_3| - |\tau_2/r_2 - \tau_3/r_3|)}{2} + \delta_3, \\ \xi_3 &= O^-(\tau_2/r_2 - \tau_3/r_3) + \frac{O^+(\tau_1/r_1 + |\tau_3/r_3| - |\tau_2/r_2 - \tau_3/r_3|)}{2} + \delta_3, \\ \xi_4 &= O^+(\tau_2/r_2 - \tau_3/r_3) + \frac{O^+(\tau_1/r_1 + |\tau_3/r_3| - |\tau_2/r_2 - \tau_3/r_3|)}{2} + \delta_3.\end{aligned}\quad (8.61)$$

Although the bias force δ_3 can be chosen arbitrarily, its contribution on tendon forces is proportional to the vector of homogeneous solution. Figure 8.19 shows the design of a redundant force resolver in accordance with Eq. (8.61). This design can be implemented for computed torque control of the Stanford/JPL finger.

Example 8.7.3 Utah/MIT Finger Let us consider the Utah/MIT finger shown in Fig. 8.10. Assuming that the pulleys mounted at the same joint axis are of the same size (i.e., $r_1 = r_5 = r_6 = r_7$, $r_2 = r_8 = r_9$, and $r_3 = r_{10}$), the structure matrix can be written as a product of R^* and B^* , where R^* and B^* are given by Eqs. (8.30) and (8.29), respectively.

Since $n = 4$ and $m = 8$, the null space of B^* is an 8×4 matrix given by

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (8.62)$$

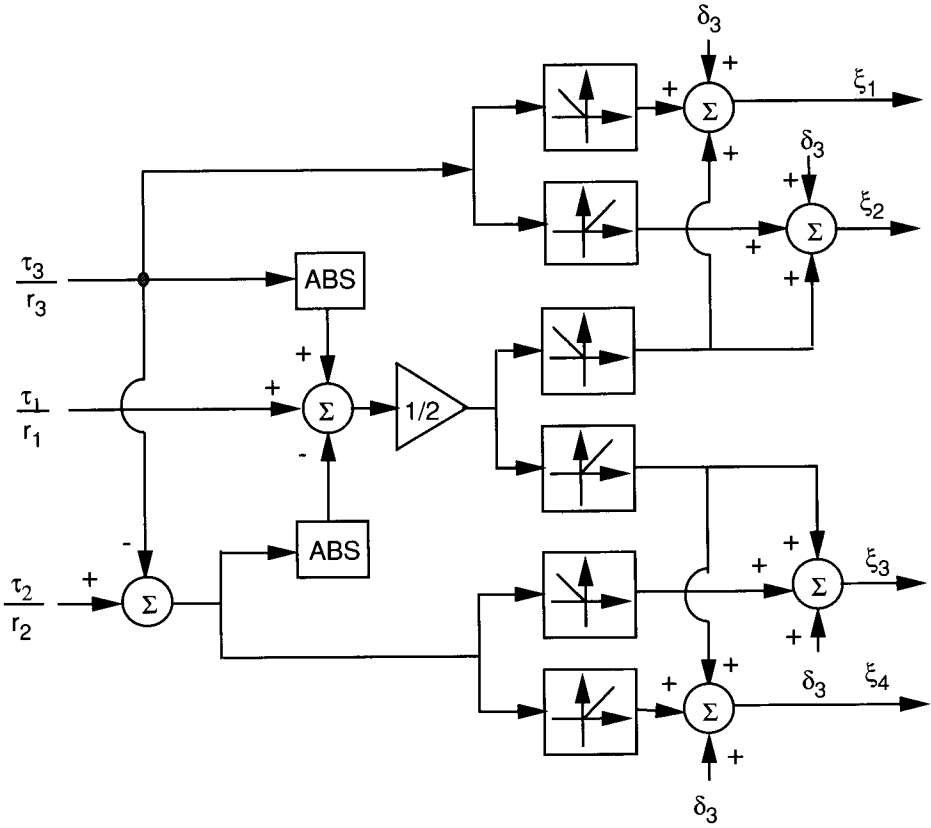


FIGURE 8.19. Force resolver schematic diagram for the Stanford/JPL finger.

Substituting $B = R^*B^*$ into Eq. (8.20) yields

$$-\xi_1 + \xi_2 - \xi_3 + \xi_4 - \xi_5 + \xi_6 - \xi_7 + \xi_8 = \tau_1/r_1, \quad (8.63)$$

$$-\xi_1 + \xi_2 - \xi_3 + \xi_4 - \xi_5 + \xi_6 = \tau_2/r_2, \quad (8.64)$$

$$-\xi_1 + \xi_2 - \xi_3 + \xi_4 = \tau_3/r_3, \quad (8.65)$$

$$-\xi_1 + \xi_2 = \tau_4/r_4. \quad (8.66)$$

Subtracting Eq. (8.64) from (8.63), (8.65) from (8.64), and (8.66) from (8.65), we obtain a simplified system of equations:

$$-\xi_7 + \xi_8 = \tau_1/r_1 - \tau_2/r_2,$$

$$-\xi_5 + \xi_6 = \tau_2/r_2 - \tau_3/r_3,$$

$$\begin{aligned}
-\xi_3 + \xi_4 &= \tau_3/r_3 - \tau_4/r_4, \\
-\xi_1 + \xi_2 &= \tau_4/r_4.
\end{aligned} \tag{8.67}$$

Since each equation in Eq. (8.67) contains only two unknowns, the four pairs of tendon forces can be written in terms of O^+ and O^- operators as follows:

$$\begin{aligned}
\xi_1 &= O^-(\tau_4/r_4) + \delta_1, \\
\xi_2 &= O^+(\tau_4/r_4) + \delta_1, \\
\xi_3 &= O^-(\tau_3/r_3 - \tau_4/r_4) + \delta_2, \\
\xi_4 &= O^+(\tau_3/r_3 - \tau_4/r_4) + \delta_2, \\
\xi_5 &= O^-(\tau_2/r_2 - \tau_3/r_3) + \delta_3, \\
\xi_6 &= O^+(\tau_2/r_2 - \tau_3/r_3) + \delta_3, \\
\xi_7 &= O^-(\tau_1/r_1 - \tau_2/r_2) + \delta_4, \\
\xi_8 &= O^+(\tau_1/r_1 - \tau_2/r_2) + \delta_4,
\end{aligned} \tag{8.68}$$

where $\delta_1, \delta_2, \delta_3$, and δ_4 are four positive bias forces. These four bias forces correspond to the four column vectors of H .

Figure 8.20 shows the design of a redundant force resolver in accordance with Eq. (8.68). In comparison with the two preceding examples, we observe that there are four bias forces for the Utah/MIT finger but only one for each of the Stanford/JPL finger and the 3×4 manipulator with a pseudotriangular structure matrix. Both the Stanford/JPL finger and the 3×4 manipulator utilize the minimum number of tendons ($m = n + 1$), whereas the Utah/MIT finger employs the maximum number of tendons ($m = 2n$). Although the Utah/MIT finger requires more tendons, it has the advantage of independently regulating the bias force for each pair of tendons. In addition, the computation of tendon forces for the Utah/MIT finger is greatly simplified.

8.8 SUMMARY

This chapter has covered the kinematics and static force analysis of tendon-driven manipulators. Tendon-driven manipulators were classified into closed-loop tendon drives and open-ended tendon drives. A planar schematic representation of tendon routing topology was defined. The fundamental circuit equations, coaxiality conditions, and the concept of transmission lines were introduced. It was also shown that tendon displacements are related to the joint angles by a structure matrix and that using the concept of transmission lines, the structure matrix can be derived by an inspection of the tendon

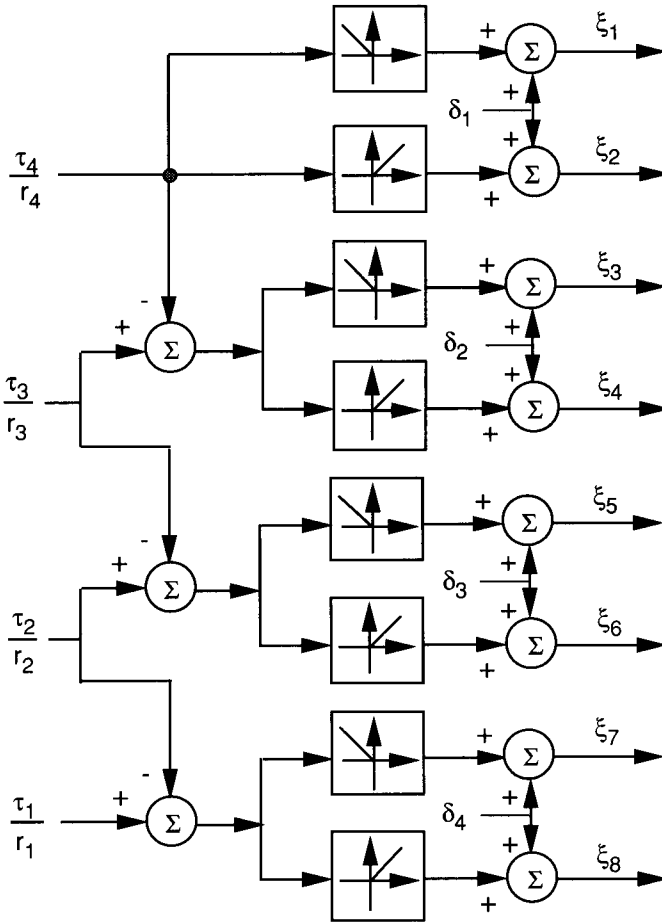


FIGURE 8.20. Force resolver schematic diagram for the Utah/MIT finger.

routing topology. From the static force analysis, structure characteristics associated with tendon-driven manipulators were identified. An atlas of tendon-driven manipulators with $n \times (n + 1)$ pseudotriangular structural matrices was developed. Finally, an efficient method for the resolution of redundant tendon forces was presented. To demonstrate the methodology, the Stanford/JPL finger and Utah/MIT finger were analyzed.

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EXERCISES

1. What is the structure matrix of the 2-dof manipulator shown in Fig. 8.2?
2. Assuming that the mass of the moving platform is 100 kg and the gravitational force points down vertically, derive the structure matrix for the 6-dof NIST RoboCrane shown in Fig. 8.21. Can the end effector resist an external force in excess of 100 kg acting in the direction opposite to gravity?
3. Derive the structure matrices for the 4-dof manipulators shown in Fig. 8.13a and b.
4. Derive the structure matrices for the 4-dof manipulators shown in Fig. 8.14h and k.
5. Are the two tendon routings shown in Fig. 8.22a and b isomorphic?

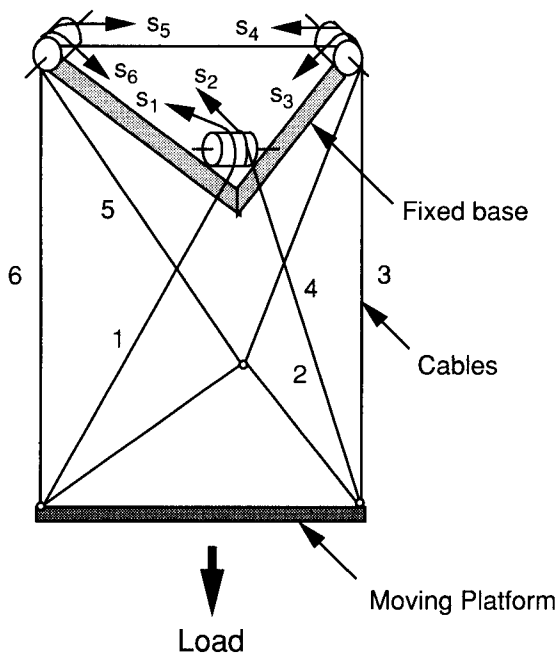


FIGURE 8.21. Functional schematic diagram of the NIST RoboCrane.

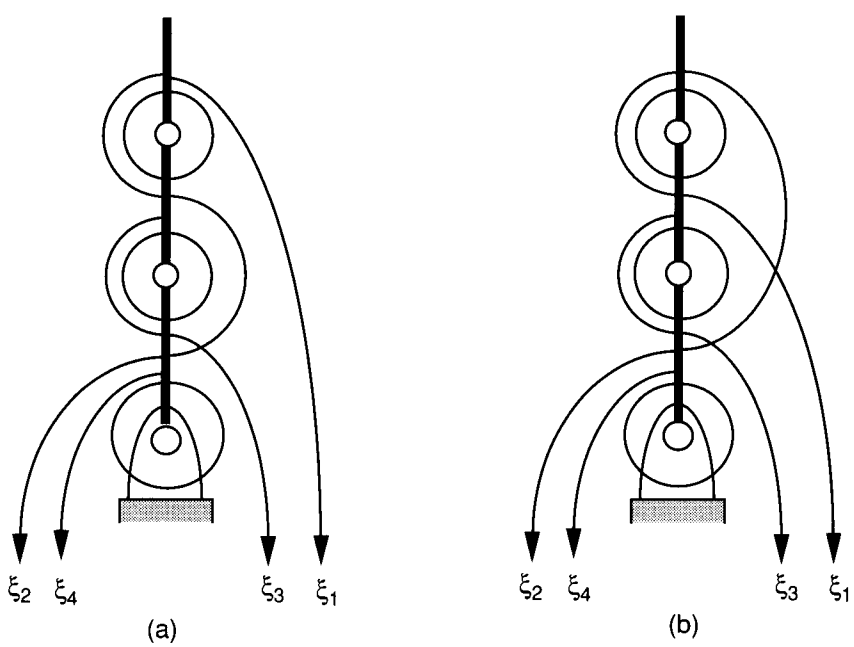


FIGURE 8.22. Two 3-dof tendon-driven manipulators with four tendons—1.

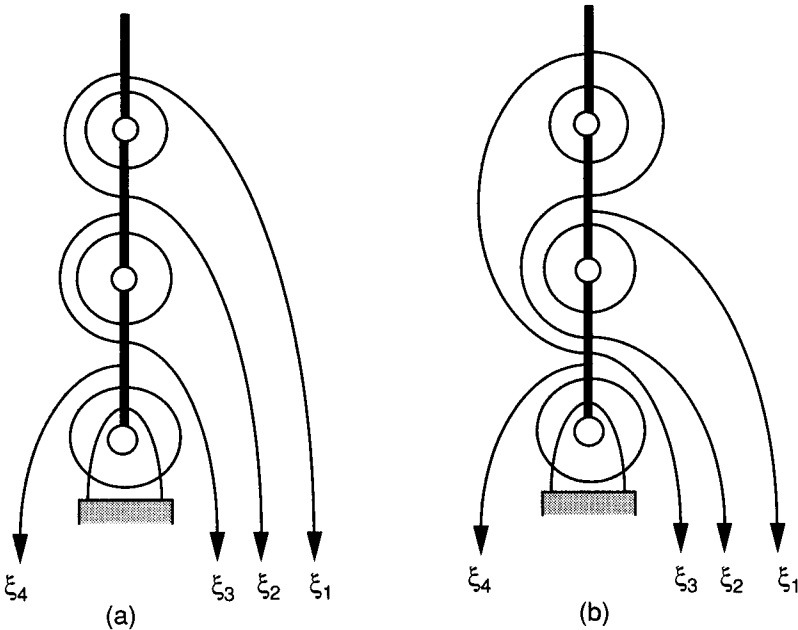


FIGURE 8.23. Two 3-dof tendon-driven manipulators with four tendons—2.

6. Prove that the two tendon routings shown in Fig. 8.23a and b are isomorphic.
7. For the 2-dof manipulator shown in Fig. 8.2, let $OA = 1$ m, $AB = 0.707$ m, $\theta_1 = 0$, and $\theta_2 = 3\pi/4$. Also let the radii of all pulleys be equal to 0.1 m. Calculate the tendon forces, (ξ_1, ξ_2, ξ_3) , required to generate an end-effector output force of $|\mathbf{f}| = 100$ N. Plot the force as a function of the orientation angle of $\mathbf{f}$.
8. For the 3-dof manipulator shown in Fig. 8.18, let the radii of all pulleys be equal to 1 unit. Calculate the tendon forces, $(\xi_1, \xi_2, \xi_3, \xi_4)$, required to produce an output force of $\mathbf{f} = (100, 0, 0)$ N at the end effector assuming that $\theta_1 = \theta_2 = 0^\circ$ and $\theta_3 = 45^\circ$.
9. Using the rectifier concept, design a tendon force resolver for the 2-dof manipulator shown in Fig. 8.2.
10. Using the rectifier concept, design a tendon force resolver for the 3-dof manipulator shown in Fig. 8.22a. Compare the result with that for the 3×4 manipulator shown in Fig. 8.18. Can you distinguish which is the better design?